Information Systems Study Guide

Simplified Notes + Exam Question Bank

🗄️ Databases | ⚖️ Ethics & Legal Issues | 💻 Software
🖥️ Hardware | 🌐 Networking & the Internet | 🛒 E-Commerce & E-Business

Built from your uploaded lecture notes, slides and textbook extracts

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> **How to use this guide:** Each topic is explained in plain, everyday language with tables and memory tips. The last two sections show you the different *styles* of questions lecturers use (definitions, MCQs, compare/contrast, essays, case studies) so nothing catches you off guard, plus a full bank of practice questions with model answers.

SECTION 1

Databases

1.1 What is a Database?

Think of a database as a **digital filing cabinet** — a well-organized collection of related data stored electronically so it's easy to find, update, and share.

> **Why we use them:** less duplication, faster searching, many people can use the data at once, and the data can be kept safe and accurate.

Real-life examples	What's stored
Banking system	Accounts, balances, transactions
Hospital records	Patients, prescriptions, appointments
School system	Students, grades, timetables
Library	Books, borrowers, due dates

1.2 DBMS — the "Manager" of the Database

A **Database Management System (DBMS)** is the software that sits between you and the raw data. Think of it as the librarian who knows exactly where everything is and lets you borrow, add, or update books safely.

> **What a DBMS does:** creates databases/tables, stores & retrieves data, allows Insert/Update/Delete, controls security & access, keeps data accurate, cuts down repeated data, lets many users share it, and backs data up.

Popular DBMS software: MySQL, Oracle, Microsoft SQL Server, PostgreSQL, Microsoft Access.

1.3 Relational vs Flat File Databases

Feature	Relational Database	Flat File Database
Structure	Many linked tables	One single table/file
Relationships	Tables connected with keys	None
Duplication	Very little	A lot
Accuracy	High	Low
Best for	Large, complex systems (e.g. a bank)	Small systems (e.g. a spreadsheet)

> **Easy way to remember:** Flat file = one big spreadsheet. Relational = several spreadsheets that "talk" to each other using shared ID numbers (keys).

1.4 The Data Hierarchy (small → big)

Think of it like sentences: letters → words → sentences → paragraphs → a book → a library.

Level	Meaning	Example
1. Bit	Smallest unit — a 0 or 1	1
2. Byte (Character)	A group of bits = one character	"A"
3. Field	One piece of information	Student Name
4. Record	Group of related fields	One student's full details
5. File (Table)	Group of related records	The Students table
6. Database	Group of related tables	The whole School Database

1.5 Common Data Types

Data Type	Stores	Example
Integer (INT)	Whole numbers	10, 1000

Decimal / Float	Numbers with decimals	45.75
Character/Text (VARCHAR, CHAR)	Letters & symbols	Names, addresses
Date	Dates	2026-05-14
Time	Time values	14:30
Boolean	TRUE / FALSE	Yes/No
Currency	Money values	$45.00

1.6 Steps to Build a Simple Relational Database

*Mnemonic: **P**lease **E**at **A** **T**asty **P**izza, **E**ven **N**ow, **C**ook **P**izza, **R**etrieve — (Purpose, Entities, Attributes, Tables, Primary key, Establish relations, Normalize, Create, Populate, Retrieve)*

1. **Identify the Purpose** — what should the system store? (e.g. bank customer management)
2. **Identify Entities** — real-world "things" e.g. Customer, Account, Employee
3. **Identify Attributes** — details that describe an entity e.g. Name, Address, Phone
4. **Define Tables** — one table per entity
5. **Assign Primary Keys** — a unique ID for each record e.g. customer_id
6. **Establish Relationships** — link tables using foreign keys
7. **Normalize** — remove duplicate data, organize efficiently
8. **Create the Database** — using SQL (CREATE DATABASE, CREATE TABLE)
9. **Populate Tables** — add data (INSERT)
10. **Retrieve Data** — SQL queries (SELECT, WHERE, JOIN, GROUP BY)

1.7 Advantages of the Database Approach

- Less duplicated data
- Better consistency (update once, applies everywhere)
- Data sharing among many users
- Stronger security via access permissions
- Better data integrity (accuracy rules)
- Backup & recovery after a crash
- Faster retrieval through queries
- Better decisions from accurate, timely info

SECTION 2

Ethics & Legal Issues in Information Systems

2.1 What Is Ethics?

Ethics = the rules of right and wrong that guide how people choose to behave, when they are free to choose. New technology is like **a rock thrown into a calm pond** — it creates ripples (new situations) that old social rules and laws haven't caught up with yet, so you're sometimes forced to make judgment calls in a "gray area."

2.2 Why Ethics Matter in Information Systems

- Protects people's privacy
- Encourages honesty and fairness
- Builds trust in technology
- Prevents harm from misuse of data/systems

2.3 The Five Moral Dimensions of the Information Age

*Mnemonic: "**I P A S Q**" — Information rights, Property rights, Accountability, System quality, Quality of life.*

Dimension	Key Question
Information rights & obligations	What can individuals/organizations protect about themselves?
Property rights & obligations	How is intellectual property protected when copying is so easy?
Accountability & control	Who is held responsible for harm done using IT?
System quality	What standard of accuracy/reliability should systems meet?
Quality of life	What values/institutions should we protect from being damaged by tech?

2.4 Key Technology Trends That Raise Ethical Issues

Trend	Why it's a problem
Computing power keeps doubling	More reliance on systems → more damage if they fail
Storage keeps getting cheaper	Easy & cheap to keep huge databases on individuals
Data analysis is more powerful	Companies can build detailed profiles of you ("profiling")
Networking/Internet advances	Data can be copied & accessed from anywhere, easily
Mobile devices	Phones can track your location without you knowing

> **Profiling** = combining data from many sources to build a detailed digital dossier on someone. **NORA (Nonobvious Relationship Awareness)** = software that finds hidden connections between people/records (e.g. spotting that a job applicant shares a phone number with a known criminal).

2.5 Responsibility, Accountability, Liability, Due Process

Term	Simple meaning
Responsibility	You accept the consequences of your own decisions
Accountability	There's a way to figure out *who* did what
Liability	Laws let a harmed person recover damages from whoever caused it
Due process	Laws are applied fairly, and you can appeal to a higher authority

2.6 Five Steps of Ethical Analysis

1. Get the facts straight — who did what, to whom, when, how
2. Define the conflict and the higher values involved (e.g. privacy vs. safety)
3. Identify the stakeholders (everyone affected)
4. Identify realistic options
5. Weigh the consequences of each option ("What if I did this every time?")

2.7 Six Candidate Ethical Principles

Principle	In plain English
Golden Rule	Treat others how you'd want to be treated
Kant's Categorical Imperative	If it's wrong for everyone to do it, it's wrong for you too
Descartes' Rule of Change	"Slippery slope" — if it can't be done repeatedly without disaster, don't start
Utilitarian Principle	Choose the action giving the greatest good/value
Risk Aversion Principle	Choose the action with the least possible harm
Ethical "No Free Lunch" Rule	Assume everything belongs to someone — give credit/compensation

2.8 Privacy & Information Rights

Privacy = the right to be left alone, free from surveillance.

> **Fair Information Practices (FIP)** — 5 principles: (1) Notice/Awareness, (2) Choice/Consent, (3) Access/Participation, (4) Security, (5) Enforcement.

Term	Meaning
Cookies	Small text files that remember you when you revisit a website
Web beacons ("web bugs")	Invisible trackers embedded in pages/emails that report your activity
Spyware	Secretly installed software that reports what you do online
Opt-in	Company can't collect data unless you specifically say yes
Opt-out	Company collects data by default, unless you say stop
Informed consent	Agreeing to something after knowing all the relevant facts
Safe harbor	A voluntary self-regulation policy that meets legal privacy standards

2.9 Property Rights: Protecting Intellectual Property

Protection	Protects	How long
Trade secret	Business ideas/formulas kept confidential	As long as it's kept secret
Copyright	The *expression* of an idea (books, software, music)	Life of author + 70 yrs (95 yrs for companies)
Patent	The underlying *idea/invention* itself	20 years

> **Remember:** Copyright protects *how* something is expressed, not the idea behind it. Patent protects the *idea* itself — much stronger, but harder to get.

Digital Millennium Copyright Act (DMCA): makes it illegal to bypass copyright protection technology; ISPs must take down pirated content once notified.

2.10 Accountability, Liability & System Quality

- It's generally **hard to hold software makers liable** like a "book" (freedom of expression), but liability grows when software acts as a **service** people depend on (e.g. ATMs).
- Three main causes of poor system performance: **software bugs, hardware/facility failure, poor data quality.**

2.11 Quality of Life Issues

Issue	Explanation
Power balance	A few big tech companies now hold huge amounts of personal data
Rapid change	"Just-in-time" jobs — businesses/workers may not have time to adapt
Boundaries (work/family/leisure)	"Anywhere computing" blurs work and personal time
Dependence & vulnerability	Society relies heavily on systems that could fail with no backup
Computer crime & abuse	Hacking, spam, malware, fraud
Digital divide	Unequal access to computers/Internet along income/race lines
Health risks	RSI, Carpal Tunnel Syndrome, Computer Vision Syndrome, Technostress

Computer crime = illegal acts done through/against a computer (e.g. hacking).
Computer abuse = unethical (but not always illegal) acts, e.g. spamming.
Spam = unsolicited junk email sent to mass audiences.

2.12 Key Legal Concepts & Laws

Law/Concept	What it does
Intellectual Property (IP)	Protects digital creations e.g. software, music
Data Protection Laws	Protect personal data from misuse
Computer Misuse Laws	Punish hacking, viruses, unauthorized access
HIPAA (1996, USA)	Protects medical/health record privacy
Gramm-Leach-Bliley Act (1999)	Privacy protection for financial services
COPPA (1998)	Requires parental consent to collect data from children under 13
CAN-SPAM Act (2003)	Bans deceptive email practices (not spam itself)
EU Data Protection Directive	Much stricter than the US — requires informed consent before use

2.13 Examples of Unethical or Illegal Behaviour

- Hacking into someone's account
- Copying software without paying (piracy)
- Spreading fake news on purpose
- Using customer data without consent
- Monitoring employees excessively without a clear policy

2.14 How to Promote Ethics & Legal Compliance

- Follow laws and workplace policies
- Get proper consent before collecting personal data
- Avoid sharing unverified information
- Respect copyright and intellectual property
- Report unethical behaviour when you see it

SECTION 3

Software Concepts

3.1 What is Software?

Software = the instructions/programs that tell a computer what to do. It's intangible (can't be touched) — unlike hardware.

> **Analogy:** Hardware is the human body; software is the mind. The body can move, but without the mind's instructions, nothing purposeful happens.

3.2 Two Main Categories of Software

Type	Job	Examples
System Software	Manages the hardware & runs the computer itself (the "factory manager")	Operating Systems (Windows, macOS, Linux), device drivers, utility programs (antivirus, disk cleanup)
Application Software	Helps the user do specific real-world tasks (the "subjects taught in class")	Word processors, spreadsheets, browsers, games

3.3 How Instructions Run

Instructions travel: **Application software → System software → Hardware** (as machine code).

Instruction type	Meaning
Sequential	Steps run one after another
Conditional	"If this, then that" — depends on a value (an "if statement")
Iterative	Repeats itself — a "loop"

3.4 Cloud Computing

Delivering computing services (storage, software, processing) **over the Internet** instead of on your own computer.

> **Analogy:** Like electricity — you don't generate your own power at home, you plug into a shared grid and pay for what you use.

Advantages	Disadvantages
Lower cost (pay only for what's used)	Needs reliable Internet connection
Scalable (grow/shrink easily)	Privacy/security concerns (data on someone else's server)
Accessible from anywhere	Less control over infrastructure
Provider handles maintenance/updates	Subscription costs add up over time
Better backup & disaster recovery	—

3.5 Ways to Obtain Software

Method	Description
Commercial/purchased	Buy a license from a vendor
Subscription-based	Pay periodically for continued access/updates
Freeware	Free to use, but still copyrighted (can't edit source code)
Shareware	Try free for a limited time, then pay
Custom-built	Developed in-house or by hired developers
Open source	Source code is free to view, modify, and share

3.6 Open-Source Software

Open-source = the underlying code is public, so anyone can view, change, and share it (e.g. Ubuntu, Mozilla Firefox).

> **Key traits:** visible source code, freedom to modify, community collaboration, usually free/low-cost, more transparent (so bugs get

fixed faster).

Analogy: Open source is a shared recipe anyone can read and improve — proprietary software is a locked recipe you can't change.

3.7 Other Key Software Terms

Term	Meaning
Software licensing	Legal permission/protection controlling how software may be used
Software upgrade	A newer version with fixes and new features
Proprietary software	Owned by a company; source code is hidden (opposite of open source)

SECTION 4

Hardware Concepts

4.1 Digital Devices

A **digital device** processes information using binary code (0s and 1s) rather than continuous (analog) signals. Examples: computers, smartphones, tablets, smart TVs, ATMs, smart watches.

4.2 The Information Processing Cycle

*Mnemonic: **I** **P**ut **M**y **O**range — Input, Processing, Memory/Storage, Output.*

Component	Role	Examples
Input	Enter data/instructions (like human senses)	Keyboard, mouse, mic, scanner
CPU ("the brain")	Calculates, decides, controls everything	Control Unit, ALU, Registers
Memory (RAM)	Temporary "working desk" — cleared when power is off (volatile)	RAM, cache
Storage	Permanent "filing cabinet" — keeps data after power off	HDD, SSD, flash drive, cloud
Output	Shows results to the user	Monitor, printer, speakers

> **CPU's 3 parts:** Control Unit (CU) — directs traffic; Arithmetic Logic Unit (ALU) — does math/logic; Registers — tiny super-fast temporary storage.

4.3 What Determines a Computer's Power?

- **Processor speed** (GHz) — more GHz = faster instruction execution
- **Number of cores** — more cores = better multitasking
- **RAM size** — more RAM = handle bigger tasks without slowing down
- **Storage type** — SSD is much faster than HDD
- **GPU** — boosts graphics/AI-heavy tasks
- **Software optimization** — efficient code uses hardware better

4.4 Memory Types

Type	Nature	Examples
Primary memory	Fast, temporary	RAM, cache
Secondary memory	Slower, permanent	Hard drives, flash drives, cloud storage

4.5 Hardware Trends

Trend	Meaning
Mobile computing	Portable devices letting you work/communicate anywhere (phones, laptops, wearables)
Ubiquitous/integrating computing	Computing built invisibly into everyday objects (smart homes, smart cars)

4.6 Commoditization of Computers

Computers went from rare/expensive (owned by governments/big firms) to cheap, everyday products — thanks to mass production, competition, and standard parts.

Positive impacts	Negative impacts
Wider access, more education/business opportunities	Devices replaced quickly → more e-waste

4.7 E-Waste (Electronic Waste)

Discarded electronics (old computers, phones, TVs, batteries). Contains toxic substances (**lead, mercury, cadmium**) that harm soil, water, air, and health if dumped/burned carelessly.

> **Solutions:** recycling programs, repairing instead of replacing, refurbishing, government regulation, and manufacturers designing more sustainable products.

SECTION 5

Networking & the Internet

5.1 What is a Network?

A **network** = two or more computers connected together so they can share resources (files, printers, hard drives, CPU).

5.2 Three Modes of Data Transmission

Mode	Direction	Example
Simplex	One-way only	TV station → your TV
Half-duplex	Both ways, but one at a time	Two-way radios
Full-duplex	Both ways, at the same time	DSL/cable broadband, telephone call

5.3 LAN Topologies (layouts)

> **Physical topology** = the actual physical cable layout. **Logical topology** = how data actually travels/how hosts access the network.

Topology	Layout	Pros / Cons
Mesh	Every device connects to every other device	+ Very reliable (no single point of failure) − Expensive/hard to install. The Internet is a mesh.
Star	All devices connect to one central hub/switch/router	+ Easy to troubleshoot − Costs more cable, central point can fail
Bus	All devices share one common cable (with terminators at ends)	+ Cheap/simple − Only 1 device can transmit at a time; rarely used today
Ring	Devices connected in a circle; a "token" travels around	+ No data collisions − If cable breaks, network is affected

5.4 Internet Basics

Term	Meaning
Internet	A massive "network of networks"
World Wide Web (WWW)	A collection of linked resources/pages accessible via browsers (part of the Internet, not the same thing!)
HTML	The language used to build Web pages
HTTP	The protocol browsers use to fetch Web pages
Hyperlink	Clickable link that connects Web pages

> **Important distinction:** The Internet is the physical network of networks; the Web is just one *service* that runs on top of it (like email, FTP are others).

5.5 Client–Server vs Peer-to-Peer

Model	How it works
Client–Server	Clients request data; servers store data and respond (e.g. browsing a website)
Peer-to-Peer (P2P)	Every computer has equal role, no central control; best for 10 or fewer computers; each user shares what they choose

5.6 IP Addresses & Protocols

- **IP address** — a unique 32-bit number (4 groups of 8 bits, called "octets") identifying a device on a network, e.g. 131.123.35.4. Can be static (fixed) or dynamic (changes).
- **DHCP** — server that automatically assigns IP addresses (vs. manual configuration).
- **MAC address** — a permanent, "burned-in" hardware address on the Network Interface Card (NIC); cannot be changed (unlike the IP address).
- **Protocol** — a set of rules for how computers communicate (e.g. identifying errors, addressing data, compressing data).

Protocol	Used for

HTTP	Loading web pages
SMTP	Sending email
FTP	Transferring files
TCP/IP	The core Internet communication suite

5.7 Network Devices

Device	What it does
Hub	Repeats a signal out to ALL ports (dumb — causes lots of collisions)
Bridge	Decides whether to pass a frame to another network segment or drop it (2 ports)
Switch	"Smart" multiport bridge — uses a MAC address table to send data only where it needs to go
Router	Connects entire networks together using IP addresses; picks the best route
Wireless Access Point	Lets wireless devices join the network via radio waves (limited range)

5.8 Types of Networks (by size)

Network	Covers
PAN (Personal Area Network)	Devices near one person (e.g. Bluetooth headset)
LAN (Local Area Network)	A home, office or building
CAN (Campus Area Network)	Several LANs in one area, e.g. a college campus
MAN (Metropolitan Area Network)	A whole town/city
WAN (Wide Area Network)	Across cities/countries (the biggest WAN = the Internet)
Intranet	Private LAN for one organization only (protected by a firewall)
Extranet	Secure network connecting a company with selected outsiders (e.g. suppliers)

5.9 Physical Transmission Media (Cables)

Cable type	Notes
Twisted-pair (UTP/STP)	Most common; cheap; UTP = unshielded (common in LANs), STP = shielded (better noise protection)
Coaxial cable	Copper core with heavy shielding; used for cable TV/older networks
Fiber-optic	Glass/plastic strands carrying light; fastest, longest range, not affected by electrical interference, but most expensive

5.10 Internet Service & Web Basics

- **ISP (Internet Service Provider)** — company providing your Internet connection. Choose based on cost, speed/bandwidth, reliability, customer service, and email options.
- **URL (Uniform Resource Locator)** — a web page's unique address, made of: `protocol://domain/path` (e.g. `http://www.nytimes.com/pages/`).
- **Domain types:** .com (anyone), .edu (schools), .gov (government), .org (organizations), .net (networking orgs).
- **Search engines** (Google, Bing, Yahoo) use a "spider"/"indexer" to find and rank pages; search methods include keyword, field, and Boolean searches.

5.11 Internet Capabilities

Capability	Examples
Communicate	Email, mailing lists, newsgroups
Retrieve	Search engines, browsing
Shop, Buy & Sell	E-commerce

> **Spam** = unsolicited junk email. **How to avoid it:** don't volunteer your email, don't publish it online, use a second email for signups, never reply to spam, use a spam filter.

Plug-in = extra software components that add abilities to a bigger application (e.g. a video player plug-in in a browser).

SECTION 6

E-Commerce & E-Business

6.1 What is E-Commerce?

E-commerce (Electronic Commerce) = buying and selling goods, services, or information over the Internet. It's the "transaction" part of doing business digitally — think online shopping, online banking, booking a flight, or paying a bill online.

6.2 What is E-Business? (and how it differs from E-Commerce)

E-Business (Electronic Business) is the *bigger picture* — it's the use of digital technology and the Internet to run an **entire** business, not just buying/selling. It includes e-commerce, plus internal processes like manufacturing, HR, finance, supply chain, and communicating with partners.

> **Easy way to remember:** All e-commerce is e-business, but not all e-business is e-commerce. E-commerce = the storefront (transactions). E-business = the whole company running digitally (transactions + internal operations + partner relationships).

Aspect	E-Commerce	E-Business
Scope	Narrower — buying/selling only	Broader — the entire business run digitally
Focus	External transactions with customers	Internal + external processes (HR, production, supply chain, customers)
Example	Ordering shoes on an online store	The company also uses digital systems for inventory, payroll, supplier orders, and customer service

6.3 Types of E-Commerce (by who is transacting with whom)

Mnemonic: think of it as "Who → To Whom" (Business, Consumer, or Government).

Type	Meaning	Example
B2C — Business to Consumer	A company sells directly to individual customers	Amazon, Jumia, Netflix
B2B — Business to Business	Businesses sell/buy from other businesses	A factory ordering raw materials from a supplier online
C2C — Consumer to Consumer	Individuals sell directly to other individuals (usually via a platform)	eBay, Facebook Marketplace, OLX
C2B — Consumer to Business	Individuals offer products/services to businesses	A freelancer selling designs to a company on Fiverr/Upwork
B2G — Business to Government	Businesses supply goods/services to government	A company winning an online government tender
C2G / G2C — Citizen/Government	Citizens interacting with government services online	Paying taxes or renewing a license online
M-Commerce — Mobile Commerce	Buying/selling using mobile devices	Shopping via a mobile app, mobile money (e.g. M-Pesa)

6.4 E-Commerce Business Models

Model	What it does	Example
Portal	A gateway offering search + a wide range of content/services	Yahoo, MSN
E-tailer (online retailer)	Sells physical products directly online	Amazon, Jumia
Content Provider	Sells information/entertainment content	Netflix, Spotify, online newspapers
Transaction Broker	Processes transactions for customers (saves time/money)	Online stock trading, flight-booking sites
Market Creator	Provides a digital platform where buyers & sellers meet	eBay, Amazon Marketplace
Service Provider	Sells an online service instead of a product	Google Drive, Dropbox, cloud software (SaaS)
Community Provider	Builds an online community around shared interests	Facebook, Reddit

6.5 E-Commerce Revenue Models (how these businesses make money)

Revenue model	How money is made
Sales revenue model	Selling goods/services directly
Advertising revenue model	Selling advertising space (e.g. Google, Facebook)
Subscription revenue model	Charging a recurring fee for access (e.g. Netflix)
Freemium model	Basic service free, charge for premium features
Transaction fee model	Charging a fee/commission per transaction (e.g. PayPal, eBay)
Affiliate revenue model	Earning a commission for referring customers to another site

6.6 Advantages of E-Commerce

For Businesses	For Customers
Reaches a global market, 24/7	Convenience — shop anytime, anywhere
Lower operating costs (no physical store needed)	Wider choice/comparison of products & prices
Easier to collect customer data for marketing	Often cheaper prices
Faster transactions and order processing	Home delivery

6.7 Disadvantages / Challenges of E-Commerce

- **Security risks** — hacking, credit card fraud, identity theft
- **Lack of personal touch** — can't physically inspect or try products before buying
- **Requires Internet access** — excludes people without reliable connectivity (digital divide)
- **Delivery/logistics issues** — delays, damaged goods, returns are harder to manage
- **Trust issues** — customers may distrust unfamiliar online sellers
- **Legal/regulatory challenges** — different laws across countries (e.g. taxation, consumer protection)

6.8 Electronic Payment Systems

Payment method	How it works
Credit/Debit card payments	Card details entered and processed through a payment gateway
Digital wallets	Store card/bank info for quick checkout (e.g. Apple Pay, Google Pay)
Mobile money	Payments made via mobile phone accounts (e.g. M-Pesa, MTN Mobile Money)
E-cash / digital currency	Digital equivalent of cash, stored electronically
Cryptocurrency	Decentralized digital currency (e.g. Bitcoin) using blockchain technology
Electronic Funds Transfer (EFT)	Direct transfer of money between bank accounts online

> **Payment gateway** = the technology that securely captures and transfers payment data between the customer, merchant, and bank (e.g. checks the card is valid, encrypts the data).

6.9 Security Issues in E-Commerce

Threat	Description
Phishing	Fake emails/websites tricking users into giving up personal/financial info
Credit card fraud	Using stolen card details to make unauthorized purchases
Identity theft	Stealing someone's personal information to impersonate them
Denial of Service (DoS) attack	Flooding a website with traffic so it can't serve real customers
Malware/Spyware	Malicious software that steals data or damages systems
Man-in-the-middle attack	Intercepting communication between customer and merchant to steal data

> **Common protections:** SSL/TLS encryption (the padlock/ `https://` in your browser), firewalls, two-factor authentication (2FA), strong passwords, and secure payment gateways.

6.10 Key Components / Technologies of E-Business

System	Full name	Purpose
ERP	Enterprise Resource Planning	Integrates all core business processes (finance, HR, production) into one system
SCM	Supply Chain Management	Manages the flow of goods from supplier to customer
CRM	Customer Relationship Management	Manages interactions and relationships with customers
KM	Knowledge Management	Captures, organizes and shares an organization's knowledge/expertise
EDI	Electronic Data Interchange	Computer-to-computer exchange of business documents (e.g. invoices, orders) in a standard format

6.11 Mobile Commerce (M-Commerce) & Social Commerce

- **M-Commerce** = conducting e-commerce transactions using mobile/wireless devices (smartphones, tablets) — e.g. shopping apps, mobile banking, mobile money.
- **Social Commerce** = using social media platforms (Facebook, Instagram, TikTok) to market, promote, and sell products directly.

6.12 Benefits of E-Business to Organizations

- Improved efficiency through automated/integrated processes
- Better communication with suppliers, partners, and customers
- Real-time access to business data for faster decision-making
- Reduced costs from streamlined operations
- Wider market reach and better competitiveness

SECTION 7

How Exam Questions Are Usually Set

Lecturers in Information Systems courses tend to reuse the same handful of question *styles*. Knowing the style tells you exactly how to answer. Here's each style with a real example and model answer.

Style 1 — Definition / "What is...?"

DEFINE

Q: What is a database?

A: An organized collection of related data, stored electronically, that can be easily accessed, managed, updated and retrieved.

Style 2 — Fill in the Blank

FILL-IN

Q: A _____ grants an inventor an exclusive monopoly on an idea for 20 years.

A: Patent

Style 3 — True or False

TRUE/FALSE

Q: The Internet and the World Wide Web are the same thing. (True/False)

A: False — the Web is just one service that runs on the Internet; the Internet also carries email, FTP, etc.

Style 4 — Multiple Choice (MCQ)

MCQ

Q: Which network topology connects every device to every other device?
A) Bus B) Star C) Mesh D) Ring

A: C — Mesh

Style 5 — List / Enumerate

LIST

Q: List the five moral dimensions of the information age.

A: Information rights & obligations, Property rights & obligations, Accountability & control, System quality, Quality of life.

Style 6 — Compare / Contrast / Differentiate

COMPARE

Q: Differentiate between a relational database and a flat file database.

A: A relational database stores data in multiple linked tables using keys, has minimal redundancy and high integrity — good for large/complex systems. A flat file database stores everything in one single table/file, has high redundancy and lower integrity — only suitable for small systems.

Style 7 — Explain / Discuss (essay-type)

DISCUSS

Q: Discuss why contemporary information technology poses a challenge to individual privacy.

A: Falling storage costs let firms keep huge personal databases cheaply; advances in data analysis (profiling, NORA) let companies/governments combine scattered data into detailed personal dossiers; networking/the Internet make it easy to copy and access this data remotely; and mobile devices can track location without users' full awareness — together these outpace existing laws, which is why new regulations (FIP, GDPR-style rules, HIPAA, COPPA) keep being introduced.

Style 8 — Case Study / Scenario (apply theory to a situation)

CASE STUDY

Q: A company installs software that records every website its employees visit, without telling them. Using the five-step ethical analysis process, evaluate this decision.

A: (1) Facts: company monitors browsing secretly. (2) Conflict: employer's right to protect productivity/security vs. employee's

right to privacy. (3) Stakeholders: employer, employees, IT dept. (4) Options: monitor openly with a policy, monitor secretly, don't monitor at all. (5) Consequences: secret monitoring may be illegal (many jurisdictions require disclosure) and destroys trust; an open, agreed Acceptable Use Policy (like the Blackburn Rovers example) balances both interests better.

Style 9 — Match the Following

MATCH

Q: Match the device to its function: (1) Router (2) Switch (3) Hub — (a) sends data to ALL ports (b) connects whole networks using IP addresses (c) uses a MAC address table to send data to the right port only

A: 1-b, 2-c, 3-a

SECTION 8

Practice Question Bank (All Topics)

🗄️ Databases

1. What is a DBMS, and give two examples.

A DBMS (Database Management System) is software that lets users create, manage, and control access to a database. Examples: MySQL, Oracle, PostgreSQL, Microsoft SQL Server, Microsoft Access.

2. Arrange in order from smallest to largest: File, Byte, Bit, Record, Field, Database.

Bit → Byte → Field → Record → File → Database.

3. Why do organizations prefer relational databases over flat files as they grow?

Relational databases reduce data duplication, improve accuracy/consistency, and let multiple related tables be linked with keys — flat files become messy, repetitive and error-prone at scale.

4. What is a primary key, and why is it important?

A primary key uniquely identifies each record in a table (e.g. customer_id), preventing duplicate or ambiguous records and enabling relationships between tables.

5. List four advantages of the database approach over traditional file systems.

Any four of: reduced redundancy, improved consistency, data sharing, enhanced security, better data integrity, backup & recovery, faster retrieval, better decision-making.

⚖️ Ethics & Legal Issues

1. Define "ethics" as it applies to information systems.

Ethics is the set of principles of right and wrong that free individuals use to guide their choices and behavior when using information systems.

2. Explain the difference between copyright and patent protection.

Copyright protects the expression of an idea (e.g. the actual code/text) for the life of the author plus 70 years, but not the underlying idea. A patent protects the underlying idea/invention itself, for 20 years, and is harder to obtain.

3. What is profiling, and why is it an ethical concern?

Profiling is combining data about a person from many different sources to build a detailed digital dossier. It's concerning because it can happen without the person's knowledge or consent, exposing sensitive details of their life.

4. Using Kant's Categorical Imperative, evaluate whether it's ethical for a company to secretly sell customer data.

Kant's rule asks: "If everyone did this, could society survive?" If every company secretly sold customer data, trust in all businesses would collapse and people would stop transacting altogether — so by this principle, the action is unethical.

5. What is the difference between opt-in and opt-out privacy models?

Opt-in: a company cannot collect your data unless you actively agree first. Opt-out: a company can collect your data by default unless you actively tell them to stop.

6. Give three examples of computer crime and three examples of computer abuse.

Computer crime (illegal): hacking, identity theft, spreading malware. Computer abuse (unethical but not always illegal): spamming, excessive employee monitoring, cyberbullying.

7. Why is it difficult to hold software companies liable when their software causes harm?

Courts have historically treated software like a "book" (protected under free expression), and it's technically impossible to guarantee zero defects in complex software — but liability grows when software functions more like an essential service (e.g. ATMs, financial systems).

💻 Software

1. Differentiate between system software and application software.

System software manages the computer's hardware and provides a platform for other software to run (e.g. Windows, macOS). Application software helps users perform specific real-world tasks (e.g. MS Word, Chrome).

2. What is open-source software? Give one example.

Software whose source code is publicly available for anyone to view, modify, and share. Example: Mozilla Firefox, Ubuntu (Linux).

3. Explain cloud computing using a simple analogy.

Cloud computing delivers computing services over the Internet instead of locally — like electricity: you don't generate your own power, you connect to a shared grid and pay for what you use.

4. List two advantages and two disadvantages of cloud computing.

Advantages: lower cost, scalability. Disadvantages: dependence on internet connectivity, privacy/security concerns.

5. What is the difference between freeware and shareware?

Freeware is free to use indefinitely but remains copyrighted (you can't edit the code). Shareware can be tried free for a limited time before you must pay for a full license.

🖥 Hardware

1. Name the four stages of the information processing cycle.

Input → Processing (CPU) → Memory/Storage → Output.

2. What are the three internal parts of the CPU, and what does each do?

Control Unit (CU) — directs operations; Arithmetic Logic Unit (ALU) — performs calculations and comparisons; Registers — tiny, very fast temporary storage for data being processed.

3. Why is RAM described as "volatile"?

Because all data stored in RAM disappears once the computer loses power — it's only for temporary, active use, unlike storage devices (HDD/SSD) which keep data permanently.

4. List four factors that determine a computer's overall computing power.

Any four of: processor speed (GHz), number of cores, amount of RAM, storage type (SSD vs HDD), GPU, and software optimization.

5. What is e-waste, and why is it dangerous?

E-waste is discarded electronic devices (old computers, phones, etc). It's dangerous because they contain toxic substances (lead, mercury, cadmium) that can contaminate soil, water and air if improperly disposed of, harming health and ecosystems.

🌐 Networking

1. Why is the Internet described as "a network of networks"?

Because it's not a single network — it's a massive collection of many separate networks (businesses, governments, universities, ISPs) all interconnected and communicating using shared protocols like TCP/IP.

2. Differentiate between a static and a dynamic IP address.

A static IP address stays the same permanently. A dynamic IP address changes periodically, usually assigned automatically by a DHCP server.

3. What is the difference between an IP address and a MAC address?

An IP address is a logical address that can change (assigned by the network). A MAC address is a permanent, "burned-in" physical hardware address on the network card that cannot be changed.

4. Explain the difference between a hub and a switch.

A hub blindly repeats incoming signals out to ALL ports, causing more collisions. A switch is "smarter" — it checks a table of MAC addresses and forwards data only to the specific port that needs it.

5. List the parts of a URL using http://www.nytimes.com/pages/ as an example.

http:// = protocol; www.nytimes.com = domain name (host + top-level domain); /pages/ = path (subdirectory within the site).

6. Differentiate between a LAN, a MAN, and a WAN.

LAN = small area like a home/office/building. MAN = covers a whole town or city (connects multiple LANs). WAN = spans across cities or countries (the Internet is the largest WAN).

7. What is the difference between an intranet and an extranet?

An intranet is a private network used only within one organization, protected by a firewall. An extranet extends limited, secure access to selected outsiders (e.g. suppliers or partners).

8. Discuss common internet crimes and how they can be reduced. (Essay-style)

Common internet crimes include hacking/unauthorized access, phishing, identity theft, malware distribution, and online fraud/scams. They can be reduced through strong passwords and authentication, firewalls and antivirus software, encryption, user education/awareness, and enforcement of computer misuse laws.

🛒 E-Commerce & E-Business

1. Differentiate between e-commerce and e-business.

E-commerce is the buying and selling of goods/services over the Internet (transactions only). E-business is broader — it's the use of digital technology to run the entire business, including internal processes like HR, production and supply chain, not just sales transactions.

2. Differentiate between B2B and B2C e-commerce, with an example of each.

B2B (Business to Business) is trade between two businesses, e.g. a factory ordering raw materials from a supplier online. B2C (Business to Consumer) is a business selling directly to individual customers, e.g. buying clothes on Amazon.

3. List four e-commerce business models.

Any four of: Portal, E-tailer, Content Provider, Transaction Broker, Market Creator, Service Provider, Community Provider.

4. What is a payment gateway, and why is it important?

A payment gateway is the technology that securely captures, encrypts, and transfers payment information between the customer, merchant, and bank during an online transaction. It's important because it protects sensitive financial data and verifies that a transaction is legitimate.

5. Discuss three security threats facing e-commerce businesses and how each can be reduced. (Essay-style)

(1) Phishing — reduced through user education and email/spam filters. (2) Credit card fraud — reduced with encryption (SSL/TLS), two-factor authentication, and fraud-detection systems. (3) Denial of Service (DoS) attacks — reduced with firewalls, traffic monitoring, and having backup servers/capacity.

6. Explain the freemium revenue model with an example.

The freemium model offers a basic version of a product/service for free, while charging for premium features or upgrades — e.g. a cloud storage app that gives you free storage up to a limit, then charges for more space.

7. What is EDI, and what is it used for?

EDI (Electronic Data Interchange) is the computer-to-computer exchange of business documents (like invoices and purchase orders) in a standard electronic format, allowing businesses to trade information quickly and accurately without paper.

8. List three advantages and three disadvantages of e-commerce for a business.

Advantages: reaches a global market 24/7, lower operating costs, faster transaction processing. Disadvantages: security risks (fraud/hacking), no personal/physical interaction with customers, dependence on reliable Internet and delivery logistics.

Good luck with your exam! Go through each section once for understanding, then use Sections 6–7 to test yourself without looking at the answers first.